

Multi-Organ CT Segmentation on SegTHOR

Mid-project presentation — esophagus, heart, trachea and aorta segmentation from thoracic CT

Clinical Motivation: Organs at Risk in Thoracic CT

- [TODO] why this matters clinically

[TODO] the manual-contouring pain point, if using one

- **Task: 3D thoracic CT → esophagus, heart, trachea, aorta**

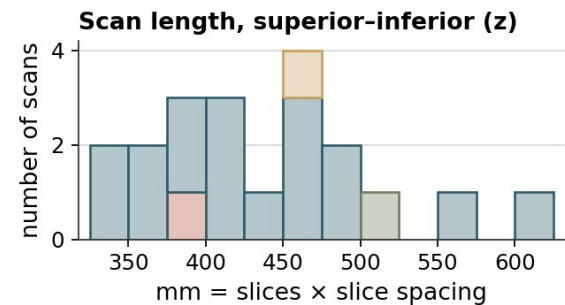
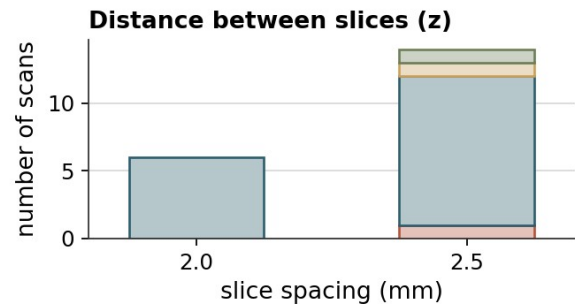
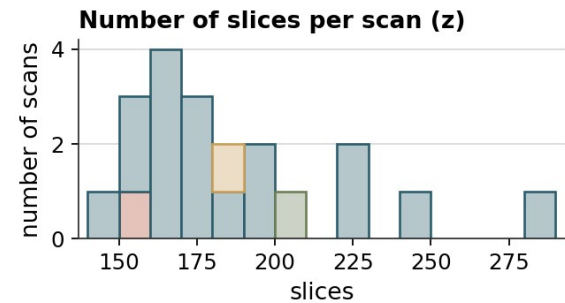
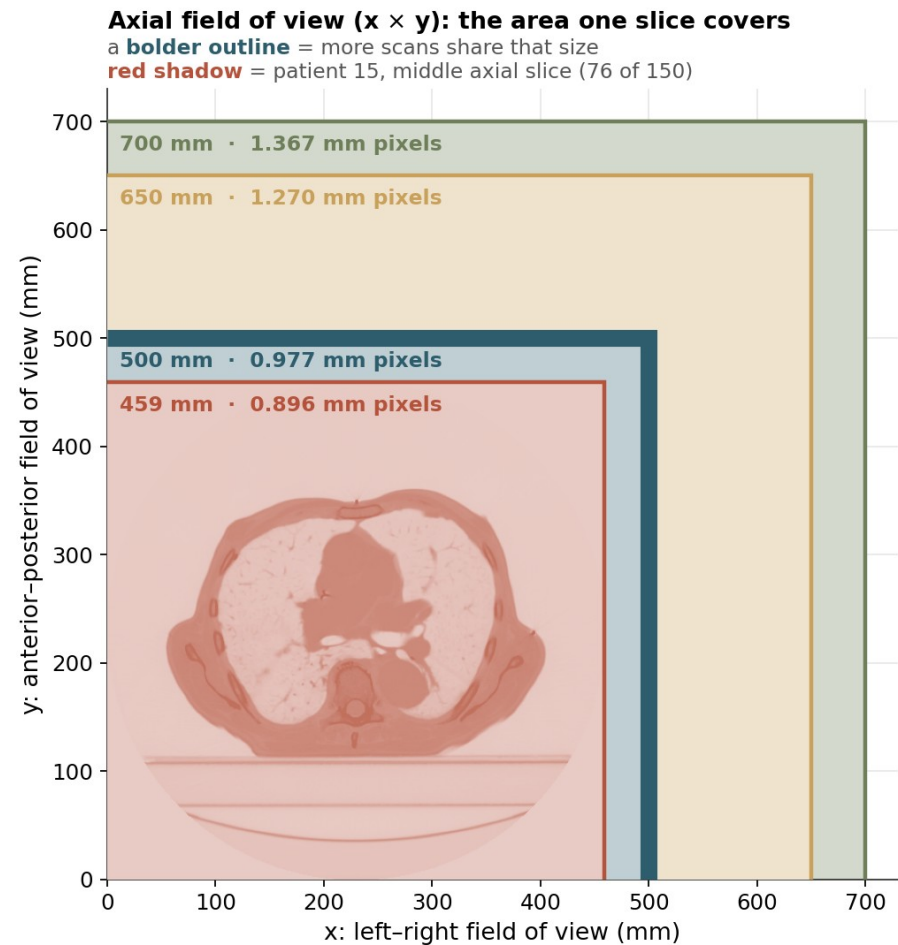
Dataset Validation: Correcting the Aorta Label

- [TODO] how the mislabeling was found and fixed
- [TODO] real per-class voxel share, now corrected — as numbers, not a bar chart
- [TODO] any other integrity checks run (e.g. alignment)

FINDING

Aorta (label 4) was mislabeled — 0 / 20 patients had it before the fix. Caught and corrected before any preprocessing or training ran on it.

Exploratory Analysis I: Scan Geometry for the 20 CT Scans



WHAT IT SHOWS

[TODO] voxel spacing across patients, slice count / scan length

HOW IT WAS MADE

[TODO] how it was measured

WHAT WE LEARN

[TODO] what this implies

Exploratory Analysis II: Whole-Scan & Per-Organ Intensity

- [TODO] whole-scan HU distribution, with each organ's voxels coloured
- [TODO] aorta vs. heart in HU
- [TODO] what this implies

Exploratory Analysis III: Label Topology & Adjacency

TOPOLOGY

- [TODO] connected components per label
- [TODO] fragmentation risk, if any

ADJACENCY

- [TODO] contact between label pairs
- [TODO] which pairs matter most, now that aorta is real

Exploratory Analysis IV: Spatial Distribution & Extent

POSITION

- [TODO] where each organ sits in 3D

EXTENT

- [TODO] coverage along the body axis

WHY THIS MATTERS

[TODO] how position and extent inform the preprocessing decisions on the next slide.

Preprocessing: From Measurement to Method

1

Step 1

[TODO], motivated by the geometry measurement
(slide 3)



2

Step 2

[TODO], motivated by the intensity measurement (slide
4)



3

Step 3

[TODO], motivated by the position/extent measurement
(slide 6)

Baseline Performance: Unmodified Pipeline (CE Loss, 25 Epochs)

- **Split: 15 train / 5 val — val = patients 01, 11, 15, 17, 19**

Scores below are means over these 5 held-out scans

POSITIVE-SLICE DICE (PATIENT MEANS)

Esophagus: 0.47

Heart: 0.58

Trachea: 0.50

Proposed Modification: Motivation, Method & Preliminary Result

1

Motivation

[TODO], tied back to an exploration finding (slides 3–6)



2

Method

[TODO] the modification tried



3

Result

[TODO] vs. baseline — from the run in progress



4

Next steps

[TODO] one or two concrete next experiments

Status & Next Steps

- Done: dataset validation, exploration, baseline, preprocessing pipeline
- In progress: [TODO] first modified-loss / modified-preprocessing run
- Next: [TODO] one or two concrete next experiments before the final deadline

Questions

Appendix A: Label Shape Variability Across Patients

- [TODO] what this figure shows, patient by patient
- [TODO] where aorta fits into that pattern

Backup only — pull in if a question opens the door